Chapter 4. Introduction to Generalized Linear Models

Exponential family, likelihood, deviance, fitting

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Generalized linear models (GLMs) contain logistic regression, Poisson loglinear models and ordinary regression. Sections 4.1–4.3 present the models; Sections 4.4–4.6 give the likelihood theory.

4.1 Generalized linear model

Motivation. For counts and proportions neither normality nor a linear mean is natural; a GLM keeps the linear predictor but chooses the response distribution and the linearizing scale separately.

4.1.1 Components of generalized linear models

Definition (Natural exponential family)

A family of the form

$$f(y_i;\theta_i) = a(\theta_i)\, b(y_i) \exp\{y_i\, Q(\theta_i)\} \tag{4.1}$$

is a *natural exponential family*; $Q(\theta)$ is the *natural parameter*.

Definition (Generalized linear model)

(i) *Random component*: independent $y_1, \ldots, y_N$, each from (4.1) (more generally, the exponential dispersion family of Section 4.4).
(ii) *Systematic component*: a *linear predictor* $\eta_i = \sum_{j=1}^{p} \beta_j x_{ij}$ (usually $x_{i1} \equiv 1$, an intercept α).
(iii) *Link function*: a monotone differentiable g with

$$g(\mu_i) = \eta_i = \sum_j \beta_j x_{ij}, \qquad \mu_i = E(Y_i). \tag{4.2}$$

The link with $g(\mu_i) = Q(\theta_i)$ is the *canonical link*; $g(\mu) = \mu$ is the *identity link*.

4.1.2 Binomial logit models for binary data

For $Y \in \{0, 1\}$ with $P(Y = 1) = \pi$, $f(y;\pi) = (1-\pi)\exp\{y \log[\pi/(1-\pi)]\}$, which is (4.1) with $a(\pi) = 1 - \pi$, $b(y) = 1$, $Q(\pi) = \operatorname{logit} \pi$. The logit is the canonical link; GLMs with it are *logit models*.

4.1.3 Poisson loglinear models for count data

For a Poisson count, $f(y;\mu) = e^{-\mu}(1/y!)\exp(y \log \mu)$, so $Q(\mu) = \log \mu$ and the canonical link is the log link:

$$\log \mu_i = \sum_j \beta_j x_{ij}, \qquad i = 1, \ldots, N, \tag{4.3}$$

the *Poisson loglinear model*.

4.1.4 Generalized linear models for continuous responses

The normal distribution (with a dispersion parameter) has the mean as natural parameter, so regression, ANOVA and ANCOVA are GLMs with identity link.

Random component	Link	Systematic	Model
Normal	identity	cont./categ./mixed	regression / ANOVA / ANCOVA
Binomial	logit	mixed	logistic regression (Ch. 5–6)
Poisson	log	mixed	loglinear (Ch. 8–9)
Multinomial	generalized logit	mixed	multinomial response (Ch. 7)

Unlike transforming Y and using least squares, the link need not stabilize variance or produce normality: fitting maximizes the likelihood of the chosen distribution.

4.1.5 Deviance

Let $L(\boldsymbol{\mu}; \mathbf{y})$ be the log likelihood in terms of the means. The *saturated model* ($\hat{\mu}_i = y_i$, one parameter per observation) attains the largest value $L(\mathbf{y}; \mathbf{y})$.

Definition (Deviance)

For a Poisson or binomial GLM, $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = -2[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})]$, the LR statistic for the model against the saturated alternative.

With N fixed and growing counts it is asymptotically χ^2_{N-p} (Theorem 4.19).

Proposition 4.1 (Deviance of the homogeneity model)

Let $y_i \sim \text{bin}(n_i, \pi_i)$ independently, $i = 1, \ldots, N$, and consider the model $\pi_i = \pi$ for all i ($p = 1$). Its deviance equals the statistic G^2 for testing independence in the $N \times 2$ table of successes and failures, and has df $= N - 1$.

Proof.

Step 1. Let $n = \sum n_i$, $s = \sum y_i$. With $L = \sum_i [y_i \log \pi_i + (n_i - y_i) \log(1 - \pi_i)]$, the MLEs are $\hat{\pi} = s/n$ (model) and $\tilde{\pi}_i = y_i/n_i$ (saturated).

Step 2. Hence

$$D = 2 \sum_i \left[y_i \log \frac{y_i}{n_i s/n} + (n_i - y_i) \log \frac{n_i - y_i}{n_i(n-s)/n} \right].$$

Step 3. The independence fitted values of the $N \times 2$ table are $n_i s/n$ and $n_i(n-s)/n$, so $D = 2 \sum \text{obs} \log(\text{obs}/\text{fit}) = G^2$, with df $= N - 1$ (N saturated parameters minus one). ■

4.1.6 Advantages of the GLM formulation

One theory and one algorithm (IRLS, Section 4.6) cover every exponential-family response and every link; deviance components serve as residuals and deviance comparisons generalize ANOVA (Section 4.5).

4.2 Generalized linear models for binary data

Motivation. For binary Y, $\text{Var}(Y) = \pi(x)[1 - \pi(x)]$, and the choice of which function of $\pi(x)$ is linear in x (identity, logit, probit) determines how effects are interpreted.

4.2.1 Linear probability model

The *linear probability model* $\pi(x) = \alpha + \beta x$ is the binomial GLM with identity link; β is the change in probability per unit x. Its defect is that $\alpha + \beta x$ eventually leaves $[0, 1]$. It should be fitted by binomial ML, not least squares, since the variance depends on x; the two agree only when fitted values stay in about 0.3–0.7.

4.2.2 Snoring and heart disease example

Example 4.2 (Snoring and heart disease)

2484 subjects, snoring scores $x = (0, 2, 4, 5)$, independent binomial rows.

Snoring	Yes	No	Proportion	Linear fit	Logit fit
Never ($x = 0$)	24	1355	0.017	0.017	0.021
Occasionally ($x = 2$)	35	603	0.055	0.057	0.044
Nearly every night ($x = 4$)	21	192	0.099	0.096	0.093
Every night ($x = 5$)	30	224	0.118	0.116	0.132

The ML linear fit is $\hat{\pi}(x) = 0.0172 + 0.0198x$ ($\widehat{\text{se}} = 0.0028$): risk 0.02 for nonsnorers, rising about 0.04 per two score units. The logistic fit is $\text{logit}\,\hat{\pi}(x) = -3.87 + 0.40x$; over this narrow range the fits are similar (Figure 1).

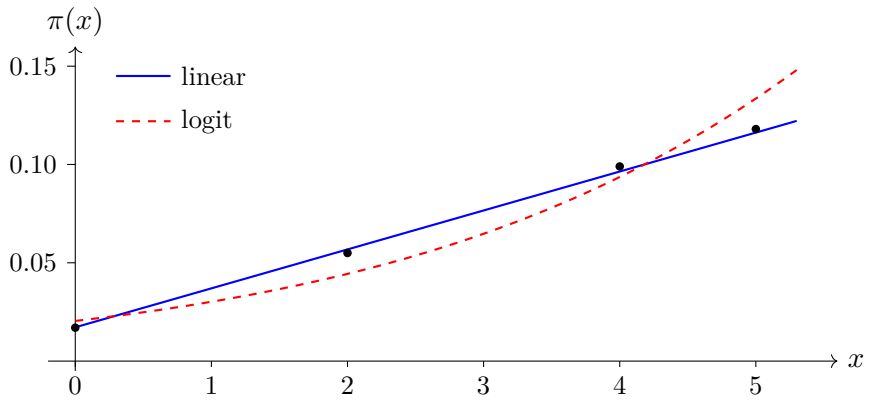


Figure 1: Snoring data: sample proportions with the linear probability and logistic fits.

4.2.3 Logistic regression model

Usually a fixed change in x matters less when $\pi(x)$ is near 0 or 1, giving S-shaped curves such as

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}. \tag{4.4}$$

Proposition 4.3 (Logistic regression is the binomial GLM with logit link)

Model (4.4) holds if and only if

$$\log \frac{\pi(x)}{1 - \pi(x)} = \alpha + \beta x.$$

Hence logistic regression is the binomial GLM with the canonical (logit) link.

Proof. From (4.4), $1-\pi(x) = 1/[1+\exp(\alpha+\beta x)]$, so the odds are $\exp(\alpha+\beta x)$; take logs. Conversely, solving $\log[\pi/(1-\pi)] = \eta$ gives $\pi = e^\eta/(1+e^\eta)$. The logit is the binomial natural parameter (Section 4.1.2). ■

The logit ranges over $\mathbb{R}$, so the structural problem of the linear model disappears; for the snoring data $\hat{\beta} = 0.40 > 0$.

4.2.4 Binomial GLM for 2×2 contingency tables

Proposition 4.4 (Effect parameters for a binary predictor)

Let $x \in \{0,1\}$ and $\text{link}[\pi(x)] = \alpha + \beta x$. Then $\beta = \text{link}[\pi(1)] - \text{link}[\pi(0)]$; in particular
(a) identity link: $\beta = \pi(1) - \pi(0)$, the difference of proportions;
(b) log link: $\beta = \log[\pi(1)/\pi(0)]$, the log relative risk;
(c) logit link: $\beta = \log \dfrac{\pi(1)/[1-\pi(1)]}{\pi(0)/[1-\pi(0)]}$, the log odds ratio.

Proof. Subtract $\text{link}[\pi(0)] = \alpha$ from $\text{link}[\pi(1)] = \alpha + \beta$ and specialize the link; for (c), a difference of log odds is a log odds ratio. ■

4.2.5 Probit and inverse cdf link functions

A monotone curve has the shape of a cdf: for a standard cdf Φ of a location–scale class,

$$\pi(x) = \Phi(\alpha+\beta x) \quad \Longleftrightarrow \quad \Phi^{-1}[\pi(x)] = \alpha + \beta x, \tag{4.5}$$

with link Φ^{-1}. The standard normal Φ gives the *probit model* (Section 6.6). The standard logistic cdf $e^x/(1+e^x)$ (sd $\pi/\sqrt{3}$, slightly heavier tails than the normal) has inverse logit π, so the logit is the inverse cdf link of the logistic distribution.

4.3 Generalized linear models for counts

Motivation. Counts are nonnegative with variance growing in the mean; the Poisson log-linear GLM is basic, and overdispersion leads to the negative binomial.

4.3.1 Poisson loglinear models

With one predictor,

$$\log \mu = \alpha + \beta x, \qquad \mu = \exp(\alpha+\beta x) = e^\alpha (e^\beta)^x, \tag{4.6}$$

so a one-unit increase in x multiplies the mean by e^β.

4.3.2 Horseshoe crab mating example

Example 4.5 (Horseshoe crab satellites)

For 173 female crabs, Y = number of satellite males; predictors color, spine condition, weight and width x (mean 26.3 cm). Grouped by width:

Width (cm)	< 23.25	-24.25	-25.25	-26.25	-27.25	-28.25	-29.25	> 29.25
Cases	14	14	28	39	22	24	18	14
Mean	1.00	1.43	2.39	2.69	2.86	3.87	3.94	5.14
Variance	2.77	8.88	6.54	11.38	6.88	8.81	16.88	8.29

(a) Log link: $\log \hat{\mu} = -3.305 + 0.164x$ ($\widehat{\text{se}} = 0.020$); $\hat{\mu} = 2.74$ at $x = 26.3$, and $e^{0.164} = 1.18$: each cm raises the mean by 18%.
(b) Identity link: $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$). The fits agree where most data lie.

4.3.3 Overdispersion for Poisson GLMs

Above, the variances far exceed the means: *overdispersion*, typically from unobserved heterogeneity.

Proposition 4.6 (Mixtures of Poissons are overdispersed)

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and λ is random with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \text{Var}(\lambda) \geq \mu$, with equality only when λ is degenerate.

Proof. [Supplement] $E(Y) = E[E(Y \mid \lambda)] = \mu$ and $\text{Var}(Y) = E[\text{Var}(Y \mid \lambda)] + \text{Var}[E(Y \mid \lambda)] = \mu + \text{Var}(\lambda)$. ■

If the mean model is right but the distribution is not Poisson, the ML estimates stay consistent but the standard errors are wrong.

4.3.4 Negative binomial GLMs

The negative binomial mass function is

$$f(y; k, \mu) = \frac{\Gamma(y+k)}{\Gamma(k)\Gamma(y+1)} \Big(\frac{k}{\mu+k}\Big)^k \Big(1 - \frac{k}{\mu+k}\Big)^y, \qquad y = 0, 1, 2, \ldots \tag{4.7}$$

Proposition 4.7 (Negative binomial as a gamma mixture)

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and λ has the gamma density with shape k and mean μ, then Y has the mass function (4.7), with $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \mu^2/k$. For fixed k, (4.7) is a natural exponential family with natural parameter $\log[\mu/(\mu+k)]$, so negative binomial regression is a GLM. As $k^{-1} \to 0$ it converges to the Poisson.

Proof. [Supplement]

Step 1. With $h(\lambda) = (k/\mu)^k \lambda^{k-1} e^{-k\lambda/\mu}/\Gamma(k)$,

$$P(Y = y) = \frac{(k/\mu)^k}{\Gamma(k)\, y!} \int_0^\infty \lambda^{y+k-1} e^{-\lambda(1+k/\mu)} d\lambda = \frac{\Gamma(y+k)}{\Gamma(k)\, y!} \Big(\frac{k}{\mu+k}\Big)^k \Big(\frac{\mu}{\mu+k}\Big)^y,$$

which is (4.7).

Step 2. The gamma law has mean μ, variance μ^2/k; Proposition 4.6 gives the moments.

Step 3. With $Q = \log[\mu/(\mu+k)]$, (4.7) is $(1 - e^Q)^k\, b(y) \exp(yQ)$, $b(y) = \Gamma(y+k)/[\Gamma(k)y!]$: the form (4.1).

Step 4. As $k \to \infty$, $\Gamma(y+k)/[\Gamma(k)(\mu+k)^y] \to 1$ and $(1 + \mu/k)^{-k} \to e^{-\mu}$, so $f(y) \to e^{-\mu}\mu^y/y!$. ■

k^{-1} is a *dispersion parameter* (Section 13.4).

Example 4.8 (Crab satellites: negative binomial fit)

Identity link: Poisson $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$); negative binomial $\hat{\mu} = -11.15 + 0.53x$ ($\widehat{\text{se}} = 0.11$), $\hat{k}^{-1} = 0.98$. Similar fits, but the honest SE is nearly doubled.

4.3.5 Poisson regression for rates

For a rate μ_i/t_i with exposure t_i, $\log(\mu_i/t_i) = \sum_j \beta_j x_{ij}$ is the Poisson GLM $\log \mu_i = \log t_i + \sum_j \beta_j x_{ij}$ with *offset* $\log t_i$ (Section 9.7).

4.3.6 Poisson GLM of independence in $I \times J$ contingency tables

Proposition 4.9 (Poisson form of independence)

Let Y_{ij} be independent Poisson with means $\mu_{ij} = \mu\alpha_i\beta_j$, where $\alpha_i, \beta_j > 0$ and $\sum_i \alpha_i = \sum_j \beta_j = 1$; equivalently

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \qquad \lambda = \log \mu,\ \lambda_i^X = \log \alpha_i,\ \lambda_j^Y = \log \beta_j. \tag{4.8}$$

Conditional on $n = \sum_{i,j} Y_{ij}$, the cell counts are multinomial with probabilities $\pi_{ij} = \alpha_i\beta_j$, which satisfy $\pi_{i+} = \alpha_i$, $\pi_{+j} = \beta_j$ and $\pi_{ij} = \pi_{i+}\pi_{+j}$.

Proof.

Step 1. $n \sim \text{Poisson}(\sum \mu_{ij}) = \text{Poisson}(\mu)$.

Step 2. For counts $\{n_{ij}\}$ with total n,

$$P(\{Y_{ij} = n_{ij}\} \mid n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!}{e^{-\mu}\mu^n/n!} = \frac{n!}{\prod_{ij} n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu}\Big)^{n_{ij}},$$

multinomial with $\pi_{ij} = \alpha_i\beta_j$.

Step 3. $\pi_{i+} = \alpha_i \sum_j \beta_j = \alpha_i$ and $\pi_{+j} = \beta_j$, so $\pi_{ij} = \pi_{i+}\pi_{+j}$. ∎

So Chapter 3 independence inferences are inferences for (4.8); Chapters 8–9 extend them.

4.4 Moments and likelihood for generalized linear models

Motivation. Mean and variance come from derivatives of one function $b(\theta)$, and the likelihood equations and information depend on the distribution only through them.

Definition (Exponential dispersion family)

Independent $y_1, \ldots, y_N$ with

$$f(y_i; \theta_i, \phi) = \exp\Big\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\Big\}; \tag{4.9}$$

θ_i is the *natural* and ϕ the *dispersion parameter*; usually $a(\phi) = \phi/\omega_i$ with known weight ω_i. For known ϕ this is (4.1) with $Q(\theta) = \theta/a(\phi)$.

4.4.1 Mean and variance functions for the random component

With $L_i = \log f(y_i; \theta_i, \phi)$ and $L = \sum_i L_i$,

$$L_i = \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi). \tag{4.10}$$

Theorem 4.10 (Moments of the exponential dispersion family)

For (4.9), with θ_i interior to the natural parameter space,

$$\mu_i = E(Y_i) = b'(\theta_i), \qquad \mathrm{Var}(Y_i) = b''(\theta_i)\, a(\phi). \tag{4.11}$$

Proof. [Supplement]

Step 1. (Bartlett identities.) Let $\ell = \log f$. For θ interior, differentiation under $\int f\, dy = 1$ is permitted (support free of θ, locally dominated derivatives), giving $E(\partial\ell/\partial\theta) = 0$; differentiating again with $\partial f/\partial\theta = f\,\partial\ell/\partial\theta$ gives $-E(\partial^2\ell/\partial\theta^2) = E(\partial\ell/\partial\theta)^2$ (sums for discrete Y).

Step 2. From (4.10), $\partial L_i/\partial\theta_i = [y_i - b'(\theta_i)]/a(\phi)$ and $\partial^2 L_i/\partial\theta_i^2 = -b''(\theta_i)/a(\phi)$.

Step 3. The first identity gives $\mu_i = b'(\theta_i)$; the second gives $b''(\theta_i)/a(\phi) = \mathrm{Var}(Y_i)/a(\phi)^2$, i.e. $\mathrm{Var}(Y_i) = b''(\theta_i)a(\phi)$. ■

Since $b'' > 0$, b' is strictly increasing, so θ_i and μ_i determine each other.

4.4.2 Mean and variance functions for Poisson and binomial

Proposition 4.11 (Poisson and binomial in dispersion form)

(a) Poisson: $\theta_i = \log\mu_i$, $b(\theta_i) = e^{\theta_i}$, $a(\phi) = 1$, $c = -\log y_i!$; then $E(Y_i) = \mathrm{Var}(Y_i) = \mu_i$.
(b) Binomial proportion: if $n_iY_i \sim \mathrm{bin}(n_i, \pi_i)$, then $\theta_i = \log[\pi_i/(1-\pi_i)]$, $b(\theta_i) = \log(1 + e^{\theta_i})$, $a(\phi) = 1/n_i$, $c = \log\binom{n_i}{n_iy_i}$; then $E(Y_i) = \pi_i$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$.

Proof.

(a) $e^{-\mu_i}\mu_i^{y_i}/y_i! = \exp[y_i\theta_i - e^{\theta_i} - \log y_i!]$, and $b'(\theta_i) = b''(\theta_i) = e^{\theta_i} = \mu_i$ (Theorem 4.10).

(b) Since $\log(1-\pi_i) = -\log(1+e^{\theta_i})$,

$$\binom{n_i}{n_iy_i}\pi_i^{n_iy_i}(1-\pi_i)^{n_i-n_iy_i} = \exp\Big[\frac{y_i\theta_i - \log(1+e^{\theta_i})}{1/n_i} + \log\binom{n_i}{n_iy_i}\Big].$$

Then $b'(\theta_i) = e^{\theta_i}/(1+e^{\theta_i}) = \pi_i$ and $b''(\theta_i)a(\phi) = \pi_i(1-\pi_i)/n_i$. ■

For $N(\mu_i, \sigma^2)$: $\theta_i = \mu_i$, $b(\theta) = \theta^2/2$, $a(\phi) = \sigma^2$.

4.4.3 Systematic component and link function

In matrix form $\boldsymbol{\eta} = X\boldsymbol{\beta}$ with $N \times p$ *model matrix* X, and $\eta_i = g(\mu_i)$.

Proposition 4.12 (Canonical link)

The canonical link (the g with $g(\mu_i) = \theta_i$) is $g = (b')^{-1}$. For the Poisson it is the log link; for the binomial it is the logit link.

Proof. By Theorem 4.10, $\mu_i = b'(\theta_i)$ with b' strictly increasing, so $\theta_i = (b')^{-1}(\mu_i)$. Poisson: $b' = \exp$, inverse log. Binomial: $b'(\theta) = e^\theta/(1+e^\theta)$, inverse logit. ■

4.4.4 Likelihood equations for a GLM

The log likelihood

$$L(\boldsymbol{\beta}) = \sum_i \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + \sum_i c(y_i, \phi) \tag{4.12}$$

depends on $\boldsymbol{\beta}$ through $\theta_i(\mu_i)$, $\mu_i = g^{-1}(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$.

Theorem 4.13 (GLM likelihood equations)

For a GLM with exponential dispersion random component,

$$\frac{\partial L_i}{\partial \beta_j} = \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i}, \tag{4.13}$$

and the likelihood equations are

$$\sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} = 0, \qquad j = 1, \ldots, p. \tag{4.14}$$

Proof.

Step 1. By the chain rule, $\partial L_i/\partial\beta_j = (\partial L_i/\partial\theta_i)(\partial\theta_i/\partial\mu_i)(\partial\mu_i/\partial\eta_i)(\partial\eta_i/\partial\beta_j)$.

Step 2. $\partial L_i/\partial\theta_i = (y_i - \mu_i)/a(\phi)$ (proof of Theorem 4.10); $\partial\mu_i/\partial\theta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; $\partial\eta_i/\partial\beta_j = x_{ij}$.

Step 3. Multiplying, $a(\phi)$ cancels, giving (4.13); summing over i gives (4.14). ■

The equations depend on the distribution only through μ_i and the *variance function* $v(\mu_i) = \mathrm{Var}(Y_i)$: μ (Poisson), $\mu(1-\mu)$ (Bernoulli), σ^2 (normal).

4.4.5 Likelihood equations for binomial GLMs

Corollary 4.14 (Binomial likelihood equations)

Let $n_iY_i \sim \mathrm{bin}(n_i, \pi_i)$ and $\pi_i = \Phi(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$, for a continuous cdf Φ with density ϕ. The likelihood equations are

$$\sum_i \frac{n_i(y_i - \pi_i)\,x_{ij}}{\pi_i(1-\pi_i)}\,\phi(\eta_i) = 0, \qquad j = 1, \ldots, p. \tag{4.15}$$

For the logit link they reduce to

$$\sum_i n_i(y_i - \pi_i)x_{ij} = 0, \qquad j = 1, \ldots, p. \tag{4.16}$$

Proof. Substitute $\partial\pi_i/\partial\eta_i = \phi(\eta_i)$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$ in (4.14). For the logit, $\partial\eta_i/\partial\pi_i = 1/[\pi_i(1-\pi_i)]$, so $\phi(\eta_i) = \pi_i(1-\pi_i)$ cancels the denominator. ■

4.4.6 Asymptotic covariance matrix of model parameter estimators

Theorem 4.15 (Information matrix of a GLM)

The expected information matrix $\mathcal{I}$ with entries $E[-\partial^2 L(\boldsymbol{\beta})/\partial\beta_h\partial\beta_j]$ is

$$\mathcal{I} = X^\top W X, \qquad W = \operatorname{diag}(w_1,\dots,w_N), \quad w_i = \frac{(\partial\mu_i/\partial\eta_i)^2}{\operatorname{Var}(Y_i)}. \tag{4.17}$$

Proof.

Step 1. By the second Bartlett identity applied to $\boldsymbol{\beta}$, $E(\partial^2 L_i/\partial\beta_h\partial\beta_j) = -E(\partial L_i/\partial\beta_h \cdot \partial L_i/\partial\beta_j)$.

Step 2. By (4.13) and $E(Y_i-\mu_i)^2 = \operatorname{Var}(Y_i)$, the latter equals $x_{ih}x_{ij}(\partial\mu_i/\partial\eta_i)^2/\operatorname{Var}(Y_i) = x_{ih}w_i x_{ij}$.

Step 3. Summing over independent observations gives the (h,j) entry of $X^\top W X$. ■

Hence

$$\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\mathcal{I}}^{-1} = (X^\top \hat{W} X)^{-1}, \tag{4.18}$$

justified by the next theorem.

Theorem 4.16 (Asymptotic normality of the ML estimator)

Let $\boldsymbol{\beta}_0$ be the true value, interior to the parameter space, and write $\boldsymbol{u}(\boldsymbol{\beta}) = \partial L/\partial\boldsymbol{\beta}$, $H(\boldsymbol{\beta}) = \partial^2 L/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top$, $\mathcal{I}_N(\boldsymbol{\beta}) = X^\top W(\boldsymbol{\beta}) X$. Assume:
(R1) $N^{-1}\mathcal{I}_N(\boldsymbol{\beta}_0) \to \Sigma$, positive definite;
(R2) Lindeberg condition: for each $\boldsymbol{c} \in \mathbb{R}^p$, the terms $\boldsymbol{c}^\top \partial L_i/\partial\boldsymbol{\beta}$ $(i \le N)$, normalized by $(N\boldsymbol{c}^\top\Sigma\boldsymbol{c})^{1/2}$, satisfy Lindeberg's condition;
(R3) $\sup_{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\delta} \|N^{-1}H(\boldsymbol{\beta}) + \Sigma\| \xrightarrow{p} 0$ as $N\to\infty$ and then $\delta\to 0$.
Then with probability tending to one the likelihood equations have a root $\hat{\boldsymbol{\beta}}$ with $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$, and

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0}, \Sigma^{-1}), \qquad \text{i.e.} \qquad \hat{\boldsymbol{\beta}} \approx N_p(\boldsymbol{\beta}_0, (X^\top W X)^{-1}).$$

Proof. [Supplement]

Step 1. (Score CLT.) The summands $\partial L_i/\partial\boldsymbol{\beta}$ are independent, mean $\mathbf{0}$, with covariances summing to $\mathcal{I}_N(\boldsymbol{\beta}_0)$. By (R1), (R2), Lindeberg–Feller and Cramér–Wold, $N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0},\Sigma)$; so $N^{-1}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{p} \mathbf{0}$.

Step 2. (Consistent root.) Let $\lambda > 0$ be the smallest eigenvalue of Σ, and fix small δ so that, by (R3), with probability $\to 1$ all eigenvalues of $-N^{-1}H$ are $\ge \lambda/2$ on $B_\delta = \{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\| \le \delta\}$. For $\boldsymbol{\beta} \in \partial B_\delta$, Taylor's theorem gives

$$L(\boldsymbol{\beta}) - L(\boldsymbol{\beta}_0) = \boldsymbol{u}(\boldsymbol{\beta}_0)^\top(\boldsymbol{\beta}-\boldsymbol{\beta}_0) + \tfrac{1}{2}(\boldsymbol{\beta}-\boldsymbol{\beta}_0)^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}-\boldsymbol{\beta}_0) \le N\delta\Big(\frac{\|\boldsymbol{u}(\boldsymbol{\beta}_0)\|}{N} - \frac{\lambda\delta}{4}\Big),$$

negative with probability $\to 1$ by Step 1. So the maximum of L on the compact B_δ is interior, where $\boldsymbol{u} = \mathbf{0}$; as δ is arbitrary, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$.

Step 3. (Expansion.) Coordinatewise mean value theorem: $\mathbf{0} = \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \boldsymbol{u}(\boldsymbol{\beta}_0) + \tilde{H}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)$, rows of $\tilde{H}$ evaluated at intermediate points; by consistency and (R3), $-N^{-1}\tilde{H} \xrightarrow{p} \Sigma$.

Step 4. So $\sqrt{N}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0) = (-N^{-1}\tilde{H})^{-1}N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0},\Sigma^{-1})$ by Slutsky; and $N^{-1}X^\top\hat{W}X \xrightarrow{p} \Sigma$ by continuity, justifying (4.18). ■

For grouped binomial or Poisson data the same holds with N fixed and $n_i \to \infty$.

4.4.7 Likelihood equations and covariance matrix for Poisson loglinear model

Corollary 4.17 (Poisson loglinear model)

For $\log \boldsymbol{\mu} = X\boldsymbol{\beta}$ with Poisson responses, the likelihood equations are

$$\sum_i (y_i - \mu_i) x_{ij} = 0, \qquad j = 1, \dots, p, \tag{4.19}$$

and $\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = (X^\top \hat{W} X)^{-1}$ with $\hat{W} = \mathrm{diag}(\hat{\mu}_1, \dots, \hat{\mu}_N)$.

Proof. For the log link $\partial \mu_i / \partial \eta_i = \mu_i = \mathrm{Var}(Y_i)$, so (4.14) becomes (4.19) and $w_i = \mu_i^2/\mu_i = \mu_i$. ■

4.5 Inference for generalized linear models

Motivation. Wald, score and LR methods apply to any GLM parameter; LR inference is organized through the deviance, the analogue of the residual sum of squares.

Theorem 4.18 (Wald, likelihood-ratio and score tests)

Under the conditions of Theorem 4.16, for $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ each of

$$\begin{aligned} W &= (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top \hat{\mathcal{I}}\, (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) && \text{(Wald)}, \\ \Lambda &= -2[L(\boldsymbol{\beta}_0) - L(\hat{\boldsymbol{\beta}})] && \text{(likelihood ratio)}, \\ S &= \boldsymbol{u}(\boldsymbol{\beta}_0)^\top \mathcal{I}(\boldsymbol{\beta}_0)^{-1} \boldsymbol{u}(\boldsymbol{\beta}_0) && \text{(score)} \end{aligned}$$

converges in distribution to χ^2_p, and the three differ by $o_p(1)$. For nested models $M_0 \subset M_1$, where M_0 imposes q linear restrictions on $\boldsymbol{\beta}$, $\Lambda = -2[L(\hat{\boldsymbol{\beta}}_0) - L(\hat{\boldsymbol{\beta}}_1)] \xrightarrow{d} \chi^2_q$.

Proof. [Supplement] Let $\boldsymbol{Z}_N = N^{-1/2} \boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} \boldsymbol{Z} \sim N_p(\mathbf{0}, \Sigma)$; by Theorem 4.16, $\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 1. (Score.) $S = \boldsymbol{Z}_N^\top (N^{-1}\mathcal{I}_N)^{-1} \boldsymbol{Z}_N \xrightarrow{d} \boldsymbol{Z}^\top \Sigma^{-1} \boldsymbol{Z} \sim \chi^2_p$.

Step 2. (Wald.) $W = [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)]^\top (N^{-1}\hat{\mathcal{I}}) [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)] = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 3. (LR.) Expanding about $\hat{\boldsymbol{\beta}}$, where $\boldsymbol{u} = \mathbf{0}$, $L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \tfrac{1}{2}(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})$, so by (R3) $\Lambda = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 4. (Nested.) Write M_0 as $\boldsymbol{\beta} = \boldsymbol{\beta}_0 + A\boldsymbol{\gamma}$, A of full rank $p-q$; M_0 has score $A^\top \boldsymbol{u}$ and information $A^\top \mathcal{I} A$. Step 3 for both models gives

$$\Lambda = \boldsymbol{Z}_N^\top [\Sigma^{-1} - A(A^\top \Sigma A)^{-1} A^\top] \boldsymbol{Z}_N + o_p(1).$$

With $\boldsymbol{Z} = \Sigma^{1/2}\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N_p(\mathbf{0}, I)$, the limit is $\boldsymbol{\varepsilon}^\top (I - P)\boldsymbol{\varepsilon}$ with $P = \Sigma^{1/2} A (A^\top \Sigma A)^{-1} A^\top \Sigma^{1/2}$ a projection of rank $p - q$; so it is χ^2_q. ■

Inverting the tests gives the Wald interval $\hat{\beta}_j \pm z_{\alpha/2} \widehat{\mathrm{se}}(\hat{\beta}_j)$ and the profile-likelihood interval $\{\beta_j : \Lambda(\beta_j) \le \chi^2_1(\alpha)\}$.

4.5.1 Deviance and goodness of fit

Let $\tilde{\theta}_i$ ($\tilde{\mu}_i = y_i$) and $\hat{\theta}_i$ be the saturated and fitted natural parameters. By (4.12) with $a(\phi) = \phi/\omega_i$, $-2[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})]$ equals the following divided by ϕ.

Definition (Scaled deviance and deviance)

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i \omega_i\big[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\big] \tag{4.20}$$

is the *deviance*, and D/ϕ the *scaled deviance*; they coincide for Poisson and binomial ($\phi = 1$). For the normal linear model $D = \sum_i (y_i - \hat{\mu}_i)^2$.

Theorem 4.19 (Asymptotic distribution of the deviance)

For a binomial or Poisson GLM with a fixed number N of observations whose counts grow ($n_i \to \infty$, or $\mu_i \to \infty$), if the model with p parameters holds then $D(\mathbf{y};\hat{\boldsymbol{\mu}}) \xrightarrow{d} \chi^2_{N-p}$.

Proof. [Supplement] The saturated model is a GLM with N free θ_i, and the working model is the submodel given by $N - p$ linear restrictions $\boldsymbol{\theta} \in \mathrm{col}(X)$ (canonical scale; otherwise reparametrize smoothly). D is the LR statistic for these nested models, so Theorem 4.18 (growing counts as sample size) gives χ^2_{N-p}. ■

4.5.2 Deviance for Poisson models

Proposition 4.20 (Poisson deviance)

For a Poisson GLM,

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i \big[y_i \log(y_i/\hat{\mu}_i) - y_i + \hat{\mu}_i\big]. \tag{4.21}$$

If the model has log link and an intercept, $\sum_i y_i = \sum_i \hat{\mu}_i$ and $D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i y_i \log(y_i/\hat{\mu}_i)$.

Proof. Put $\hat{\theta}_i = \log\hat{\mu}_i$, $b(\hat{\theta}_i) = \hat{\mu}_i$, $\tilde{\theta}_i = \log y_i$, $b(\tilde{\theta}_i) = y_i$, $\omega_i = 1$ in (4.20). With an intercept the first equation of (4.19) is $\sum_i (y_i - \hat{\mu}_i) = 0$, so the linear terms cancel ($0 \log 0 = 0$). ■

For a two-way table under independence this is G^2 of Chapter 3.

4.5.3 Deviance for binomial models: grouped and ungrouped data

Proposition 4.21 (Binomial deviance)

For binomial proportions y_i based on n_i trials,

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i n_i y_i \log\frac{n_i y_i}{n_i\hat{\pi}_i} + 2\sum_i (n_i - n_i y_i)\log\frac{n_i - n_i y_i}{n_i - n_i\hat{\pi}_i}, \tag{4.22}$$

that is, $D = 2\sum \text{observed} \times \log(\text{observed/fitted})$ over the $2N$ cells of successes and failures.

Proof. With $\hat{\theta}_i = \mathrm{logit}\,\hat{\pi}_i$, $b(\hat{\theta}_i) = -\log(1-\hat{\pi}_i)$, $\tilde{\theta}_i = \mathrm{logit}\,y_i$, $b(\tilde{\theta}_i) = -\log(1-y_i)$, $\omega_i = n_i$, (4.20) gives

$$D = 2\sum_i n_i\Big[y_i \log\frac{y_i}{\hat{\pi}_i} + (1-y_i)\log\frac{1-y_i}{1-\hat{\pi}_i}\Big],$$

which is (4.22). ■

Grouped (counts per setting) and ungrouped (0/1) entry give the same ML estimates but different saturated models and deviances; the χ^2 approximation holds only for grouped data with $n_i \to \infty$.

4.5.4 Likelihood-ratio model comparison using the deviance

Proposition 4.22 (Deviance difference)

Let $M_0 \subset M_1$ be nested Poisson or binomial GLMs with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$. Then $D(\mathbf{y}; \hat{\boldsymbol{\mu}}_1) \leq D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0)$ and the likelihood-ratio statistic for $H_0 : M_0$ given M_1 is

$$-2\big[L(\hat{\boldsymbol{\mu}}_0; \mathbf{y}) - L(\hat{\boldsymbol{\mu}}_1; \mathbf{y})\big] = D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0) - D(\mathbf{y}; \hat{\boldsymbol{\mu}}_1),$$

asymptotically χ^2 with df equal to the difference in numbers of parameters. With log link and intercept (Poisson) or for binomial data, it equals $2 \sum \text{observed} \times \log(\text{fitted}_1/\text{fitted}_0)$.

Proof. Maximizing over a smaller space gives $L(\hat{\boldsymbol{\mu}}_0) \leq L(\hat{\boldsymbol{\mu}}_1)$; the identity follows by adding and subtracting $L(\mathbf{y}; \mathbf{y})$; the distribution is Theorem 4.18. Subtracting (4.22) (or (4.21) without linear terms) for the two models cancels the log observed terms. ■

The saturated part cancels, so the comparison is valid for ungrouped data too.

4.5.5 Residuals for GLMs

Definition (Deviance and Pearson residuals)

With $D = \sum_i d_i$, $d_i = 2\omega_i[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)]$, the *deviance residual* is $\sqrt{d_i}\ \operatorname{sgn}(y_i - \hat{\mu}_i)$ and the *Pearson residual* is $e_i = (y_i - \hat{\mu}_i)/[\widehat{\operatorname{Var}}(Y_i)]^{1/2}$.

For independence in a two-way table $\sum e_{ij}^2 = X^2$, $\sum d_{ij} = G^2$. These are less variable than $N(0,1)$; standardizing uses the *hat matrix*

$$\text{HatM} = W^{1/2} X (X^\top W X)^{-1} X^\top W^{1/2}. \tag{4.23}$$

Proposition 4.23 (Leverages and residual covariance)

(a) HatM is symmetric and idempotent with trace p; its diagonal elements (*leverages*) satisfy $0 \leq h_i \leq 1$ and $\sum_i h_i = p$.
(b) To first order, with $V = \operatorname{Cov}(\boldsymbol{Y}) = \operatorname{diag}\{\operatorname{Var}(Y_i)\}$,

$$\operatorname{Cov}(\boldsymbol{Y} - \hat{\boldsymbol{\mu}}) \approx V^{1/2}(I - \text{HatM})V^{1/2}, \qquad \operatorname{Var}(Y_i - \hat{\mu}_i) \approx \operatorname{Var}(Y_i)(1 - h_i).$$

Hence the *standardized Pearson residual* is

$$r_i = \frac{y_i - \hat{\mu}_i}{[\widehat{\operatorname{Var}}(Y_i)(1 - \hat{h}_i)]^{1/2}} = \frac{e_i}{\sqrt{1 - \hat{h}_i}} \qquad \left(\text{Poisson: } \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1 - \hat{h}_i)}}\right). \tag{4.24}$$

Proof. [Supplement] (a) With $A = W^{1/2}X$, $\text{HatM} = A(A^\top A)^{-1}A^\top$ is symmetric, $\text{HatM}^2 = \text{HatM}$, $\operatorname{tr}\text{HatM} = \operatorname{tr} I_p = p$, and $h_i = \sum_j h_{ij}^2 \geq h_i^2$ gives $0 \leq h_i \leq 1$.

(b)

Step 1. Let $G = \operatorname{diag}(\partial\eta_i/\partial\mu_i)$; by (4.13), $\boldsymbol{u} = X^\top W G(\boldsymbol{y} - \boldsymbol{\mu})$.

Step 2. By Theorem 4.16, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \approx (X^\top W X)^{-1} X^\top W G(\boldsymbol{y} - \boldsymbol{\mu})$ and $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \approx G^{-1}X(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, so $\boldsymbol{y} - \hat{\boldsymbol{\mu}} \approx [I - G^{-1}X(X^\top W X)^{-1}X^\top W G](\boldsymbol{y} - \boldsymbol{\mu})$.

Step 3. Since $w_i = 1/(G_{ii}^2 \operatorname{Var}(Y_i))$, $V^{1/2} = G^{-1}W^{-1/2} =: M$, and the bracket is $M(I - \text{HatM})M^{-1}$.

Step 4. With $\operatorname{Cov}(\boldsymbol{y} - \boldsymbol{\mu}) = M^2$ and HatM symmetric idempotent, $\operatorname{Cov}(\boldsymbol{y} - \hat{\boldsymbol{\mu}}) \approx M(I - \text{HatM})M^{-1}M^2M^{-1}(I - \text{HatM})M = M(I - \text{HatM})M$, with ith diagonal entry $\operatorname{Var}(Y_i)(1 - h_i)$.

■

Unlike ordinary regression, leverages depend on the fit through W, so extreme predictor values need not have high leverage.

4.6 Fitting generalized linear models

Motivation. The likelihood equations are nonlinear; Newton–Raphson and Fisher scoring solve them, and Fisher scoring is iteratively reweighted least squares.

4.6.1 Newton–Raphson method

With $\boldsymbol{u}^{(t)}, H^{(t)}$ the score and Hessian at $\boldsymbol{\beta}^{(t)}$, maximizing the quadratic approximation $L(\boldsymbol{\beta}^{(t)}) + \boldsymbol{u}^{(t)\top}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)}) + \tfrac{1}{2}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)})^\top H^{(t)}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)})$ gives the *Newton–Raphson* step (Figure 2)

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - (H^{(t)})^{-1}\boldsymbol{u}^{(t)}. \tag{4.25}$$

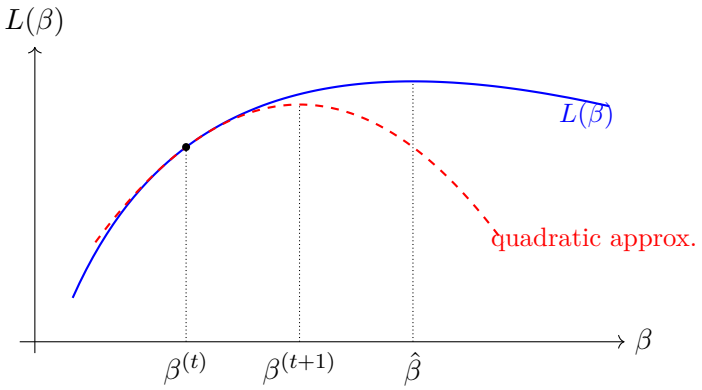


Figure 2: One Newton–Raphson cycle: the next guess maximizes the quadratic approximation at $\beta^{(t)}$.

Example 4.24 (Newton–Raphson for a binomial parameter)

For $y \sim \mathrm{bin}(n,\pi)$, $u = (y-n\pi)/[\pi(1-\pi)]$ and $H = -[y/\pi^2 + (n-y)/(1-\pi)^2]$. The step moves $\pi^{(t)}$ toward y/n; from $\pi^{(0)} = \tfrac{1}{2}$ it reaches y/n at once, otherwise in four or five iterations.

Theorem 4.25 (Quadratic convergence of Newton–Raphson)

Suppose $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, H is Lipschitz on a ball B around $\hat{\boldsymbol{\beta}}$ with constant γ ($\|H(\boldsymbol{a}) - H(\boldsymbol{b})\| \le \gamma\|\boldsymbol{a}-\boldsymbol{b}\|$), and $\|H(\boldsymbol{\beta})^{-1}\| \le M$ on B. If $\boldsymbol{\beta}^{(t)} \in B$, then

$$\|\boldsymbol{\beta}^{(t+1)} - \hat{\boldsymbol{\beta}}\| \le \frac{M\gamma}{2}\,\|\boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}\|^2.$$

In particular, if $\boldsymbol{\beta}^{(0)} \in B$ and $\|\boldsymbol{\beta}^{(0)} - \hat{\boldsymbol{\beta}}\| < 2/(M\gamma)$, the iterates converge to $\hat{\boldsymbol{\beta}}$ and the number of correct digits roughly doubles each step.

Proof. [Supplement]

Step 1. With $\boldsymbol{e}_t = \boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}$, $\boldsymbol{u}^{(t)} = \boldsymbol{u}^{(t)} - \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \int_0^1 H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)\,ds\;\boldsymbol{e}_t$.

Step 2. By (4.25),

$$\boldsymbol{e}_{t+1} = \boldsymbol{e}_t - (H^{(t)})^{-1}\boldsymbol{u}^{(t)} = (H^{(t)})^{-1}\int_0^1 [H(\hat{\boldsymbol{\beta}} + \boldsymbol{e}_t) - H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)]ds\, \boldsymbol{e}_t.$$

Step 3. The integrand has norm $\leq \gamma(1-s)\|\boldsymbol{e}_t\|$ and $\int_0^1(1-s)ds = \frac{1}{2}$, so $\|\boldsymbol{e}_{t+1}\| \leq \frac{M\gamma}{2}\|\boldsymbol{e}_t\|^2$.

Step 4. If $\rho = \frac{M\gamma}{2}\|\boldsymbol{e}_0\| < 1$, induction gives $\|\boldsymbol{e}_{t+1}\| \leq \rho\|\boldsymbol{e}_t\|$, the iterates stay in B and $\boldsymbol{e}_t \to \mathbf{0}$, the error being roughly squared each step. ■

4.6.2 Fisher scoring method

Fisher scoring replaces $-H$ (observed information) by the expected information $\mathcal{I}$:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + (\mathcal{I}^{(t)})^{-1}\boldsymbol{u}^{(t)} \quad \Longleftrightarrow \quad \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t+1)} = \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} + \boldsymbol{u}^{(t)}. \tag{4.26}$$

For the binomial parameter $\mathcal{I} = n/[\pi(1-\pi)]$ and one step gives y/n from any start. For a GLM, $\mathcal{I}^{(t)} = X^\top W^{(t)} X$ and $\boldsymbol{u}^{(t)}$ has elements

$$u_j = \sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i}; \tag{4.27}$$

at convergence $(\mathcal{I}^{(t)})^{-1}$ is the covariance estimate (4.18). For canonical links the two methods coincide (Theorem 4.27); otherwise Fisher scoring has a nonnegative definite $\mathcal{I}$ but need not converge quadratically.

4.6.3 ML as iterative reweighted least squares

For $\boldsymbol{z} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathrm{Cov}(\boldsymbol{\varepsilon}) = V$, the WLS estimator is $(X^\top V^{-1}X)^{-1}X^\top V^{-1}\boldsymbol{z}$.

Theorem 4.26 (Fisher scoring is IRLS)

For a GLM, the Fisher scoring step (4.26) is

$$\boldsymbol{\beta}^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} \boldsymbol{z}^{(t)},$$

the WLS fit of the *working response*

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)})\frac{\partial \eta_i^{(t)}}{\partial \mu_i^{(t)}} \tag{4.28}$$

on X with weight matrix (inverse covariance) $W^{(t)}$.

Proof.

Step 1. $\mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} = X^\top W^{(t)} X\boldsymbol{\beta}^{(t)} = X^\top W^{(t)}\boldsymbol{\eta}^{(t)}$.

Step 2. Since $w_i = (\partial\mu_i/\partial\eta_i)^2/\mathrm{Var}(Y_i)$, each term of (4.27) is $x_{ij}w_i(y_i - \mu_i)\,\partial\eta_i/\partial\mu_i$, so $\boldsymbol{u}^{(t)} = X^\top W^{(t)}\boldsymbol{d}^{(t)}$ with $d_i^{(t)} = (y_i - \mu_i^{(t)})\,\partial\eta_i^{(t)}/\partial\mu_i^{(t)}$.

Step 3. The right side of (4.26) is $X^\top W^{(t)}(\boldsymbol{\eta}^{(t)} + \boldsymbol{d}^{(t)}) = X^\top W^{(t)}\boldsymbol{z}^{(t)}$: the WLS normal equations. ■

z_i linearizes the link, $g(y_i) \approx \eta_i + (y_i - \mu_i)\,\partial\eta_i/\partial\mu_i$, and W^{-1} approximates its covariance. Iterating is *iteratively reweighted least squares* (IRLS); a start is $\hat{\boldsymbol{\mu}}^{(0)} = \mathbf{y}$, adjusted where $g(y_i)$ is infinite.

4.6.4 Simplifications for canonical links

Theorem 4.27 (Canonical link properties)

Consider a GLM with the canonical link, $\theta_i = \eta_i = \sum_j \beta_j x_{ij}$, and a common $a(\phi)$ for all i (e.g. Poisson; binomial with all $n_i = 1$). Then:
(a) $\sum_i y_i x_{ij}$, $j = 1, \ldots, p$, are sufficient statistics for $\boldsymbol{\beta}$;
(b) $\partial L_i / \partial \beta_j = (y_i - \mu_i) x_{ij} / a(\phi)$, so the likelihood equations are

$$\sum_i x_{ij} y_i = \sum_i x_{ij} \mu_i, \qquad j = 1, \ldots, p, \qquad \text{i.e.} \quad X^\top \mathbf{y} = X^\top \boldsymbol{\mu}; \tag{4.29}$$

(c) $\partial^2 L / \partial \beta_h \partial \beta_j = -\sum_i x_{ih} x_{ij} b''(\theta_i)/a(\phi)$ does not involve $\mathbf{y}$; hence observed and expected information coincide, $H = -\mathcal{I}$, and Newton–Raphson equals Fisher scoring;
(d) L is concave in $\boldsymbol{\beta}$, strictly if X has full column rank; so a solution of (4.29), if one exists, is the unique ML estimate.
With weights $a(\phi) = \phi/\omega_i$ the same holds with y_i, μ_i replaced by $\omega_i y_i, \omega_i \mu_i$.

Proof.

Step 1. By (4.12) the only term involving both data and parameters is $a(\phi)^{-1} \sum_i y_i \theta_i = a(\phi)^{-1} \sum_j \beta_j (\sum_i y_i x_{ij})$; the factorization theorem gives (a).

Step 2. Here $\partial \mu_i / \partial \eta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; inserting into (4.13) gives (b).

Step 3. Since $\partial \mu_i / \partial \beta_h = b''(\theta_i) x_{ih}$, $\partial^2 L_i / \partial \beta_h \partial \beta_j = -x_{ij} x_{ih} b''(\theta_i)/a(\phi)$, free of y_i and equal to its expectation: (c).

Step 4. For $\boldsymbol{c} \in \mathbb{R}^p$, $\boldsymbol{c}^\top H \boldsymbol{c} = -\sum_i b''(\theta_i)(\boldsymbol{x}_i^\top \boldsymbol{c})^2/a(\phi) \le 0$, with equality only if $X\boldsymbol{c} = \mathbf{0}$, impossible for $\boldsymbol{c} \neq \mathbf{0}$ under full rank. A stationary point of a strictly concave function is its unique maximum: (d). ■

Thus (4.29) equates sufficient statistics to their expectations: the normal equations, (4.19), and (4.16) with $n_i = 1$.